

Hardware & Networking

Detailed Chapter Notes & Worked Examples - Volume 2

This pack develops practical understanding through explanations, worked examples, lab tasks and review questions. Complete the matching Foundation Pack first.

Chapter	Topic
1	Safe diagnosis
2	Assembly and firmware
3	IP networking
4	Small office network

Chapter 1: Safe diagnosis

Concept notes

Begin with symptom, scope, recent changes and backup risk. Use ESD precautions, disconnect power and change one factor at a time.

Worked example

PC has no power: verify outlet, cable, PSU switch, front-panel connector and known-good PSU before motherboard conclusion.

Lab assignment

Create a diagnostic worksheet for five common symptoms.

Review questions

1. Why change one factor?
2. What does ESD damage?
3. When back up data?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 2: Assembly and firmware

Concept notes

Compatibility covers socket, chipset, RAM generation, power and physical size. Firmware detects hardware and controls boot settings.

Worked example

After assembly, check CPU temperature, memory size, storage detection and boot order before OS installation.

Lab assignment

Prepare compatibility list and assembly inspection record.

Review questions

1. What is POST?
2. Why verify power rating?
3. What does boot order control?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 3: IP networking

Concept notes

Devices in one subnet communicate locally; router connects networks; DNS converts names; DHCP assigns settings automatically.

Worked example

192.168.1.20/24 has network 192.168.1.0 and typical gateway 192.168.1.1.

Lab assignment

Configure two devices and test ipconfig, ping and name resolution.

Review questions

1. Purpose of gateway?
2. What does DNS do?
3. Difference switch and router?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 4: Small office network

Concept notes

Plan users, coverage, cabling, IP scheme, sharing, guest access and security. Document every device and credential owner.

Worked example

Router, 8-port switch, four PCs, printer and separate guest Wi-Fi with updated firmware.

Lab assignment

Draw topology, configure lab network and produce handover sheet.

Review questions

1. Why separate guest Wi-Fi?
2. What belongs in topology?
3. Why update firmware?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Final Applied Assignment

Create one original project that combines all four chapters of Hardware & Networking. Use realistic but non-sensitive sample data. Submit planning notes, source files, final output, test evidence and a one-page reflection.

Assessment rubric

Area	Marks
Concept accuracy	25
Working practical output	30
Validation and testing	20
File organisation and presentation	15
Original explanation	10

Student declaration

I confirm that the submitted work is my own practice. I have not included passwords, OTPs, identity documents or confidential student records.

Support: mcieducationalgroup@gmail.com | 9334779133, 7004773247